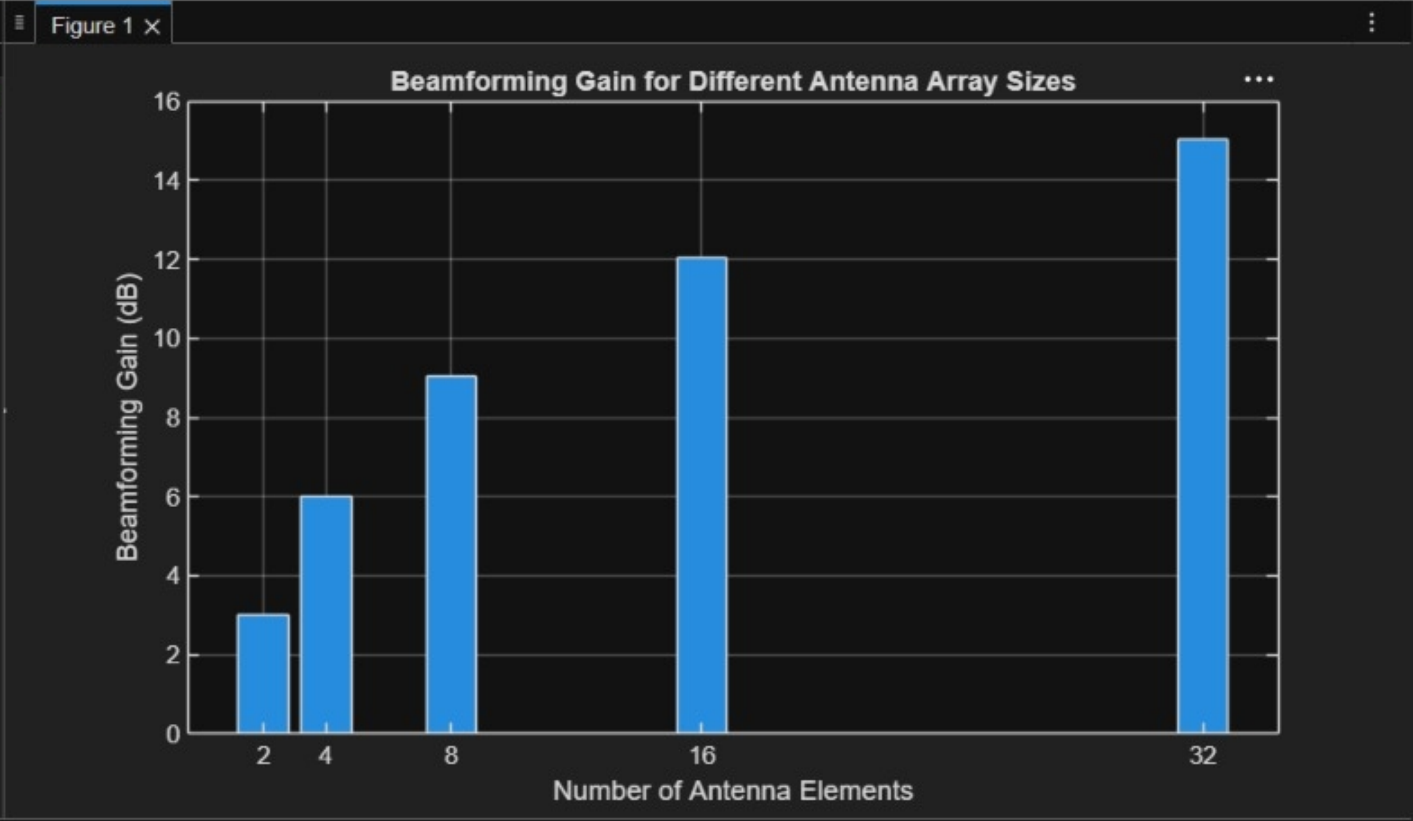


Save As, Copy Figure, Print, Show Code, FILE

TITLE LABELS AND ANNOTATIONS

Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

```
1 clc;
2 clear;
3 close all;
4 antennas = [2 4 8 16 32]; % Number of Antenna Elements
5 gain = 10*log10(antennas); % Beamforming Gain (dB)
6 figure
7 bar(antennas, gain)
8 grid on
9 xlabel('Number of Antenna Elements')
10 ylabel('Beamforming Gain (dB)')
11 title('Beamforming Gain for Different Antenna Array Sizes')
```



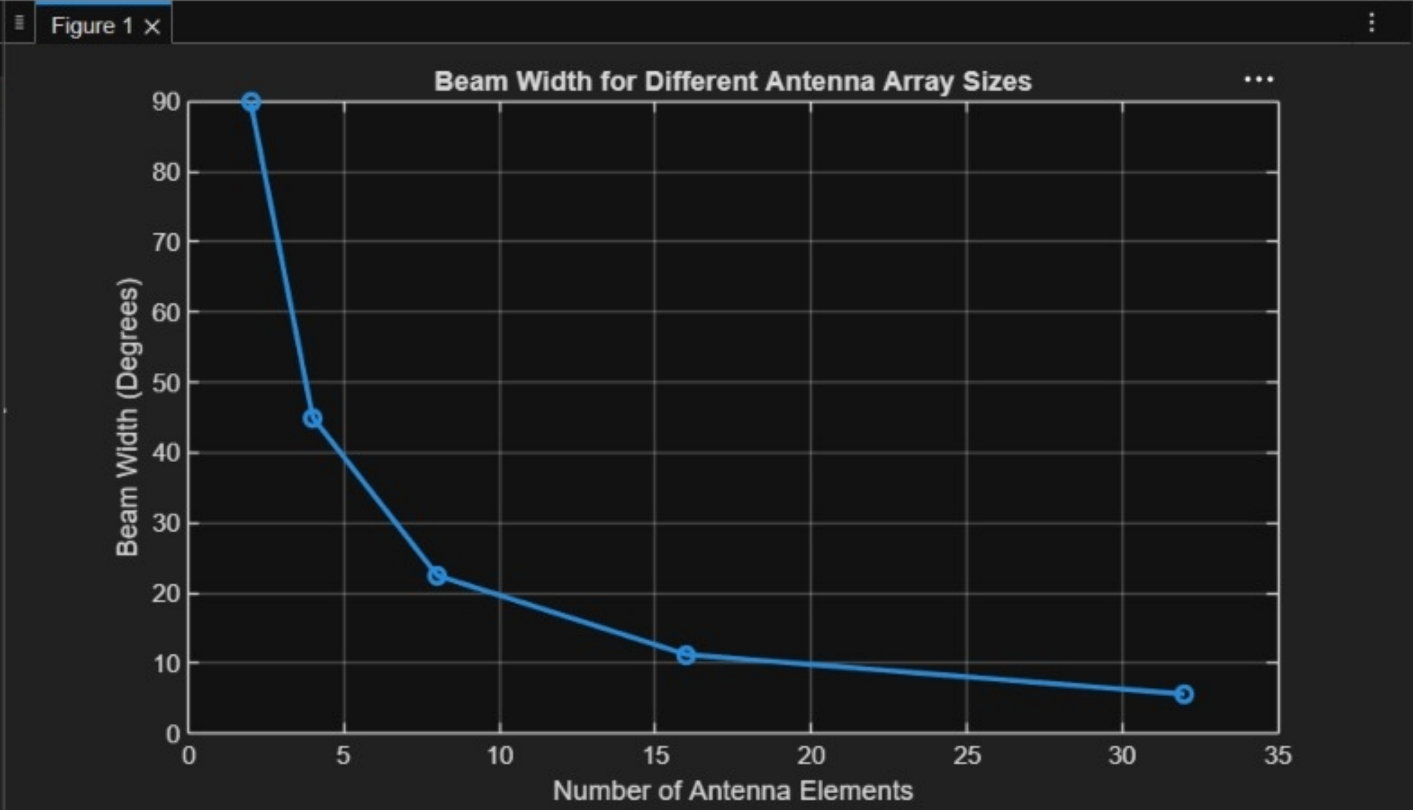
Save As, Copy Figure, Print, Show Code, FILE

TITLE AND LABELS

Title, Subtitle, X-Label, Y-Label, Z-Label, Legend, Colorbar, Grid, X-Grid, Y-Grid

LABELS AND ANNOTATIONS

```
1 clc;
2 clear;
3 close all;
4 antennas = [2 4 8 16 32]; % Number of Antenna Elements
5 beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)
6 figure
7 plot(antennas, beamwidth, 'o-', 'LineWidth', 2)
8 grid on
9 xlabel('Number of Antenna Elements')
10 ylabel('Beam Width (Degrees)')
11 title('Beam Width for Different Antenna Array Sizes')
```



Save As, Copy Figure, Print, Show Code, FILE

TITLE AND ANNOTATIONS: Title, Subtitle, X-Label, Y-Label, Z-Label, Legend, Colorbar, Grid, X-Grid, Y-Grid

```
1 clc;
2 clear;
3 close all;
4 theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
5 gain = [1 1 1 1 1]; % Constant normalized gain
6 figure
7 polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
8 title('Main Lobe Direction in Analog Beamforming')
```

